



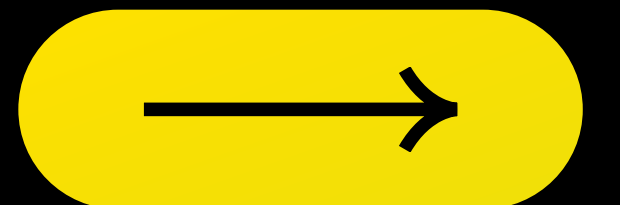
MACHINE INTELLIGENCE

# SMART SOLAR SAVINGS PREDICTOR

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**What if you could slash your electricity bills while contributing to a cleaner environment, all from the comfort of your own roof?**

# Problem Statement

Household energy costs are a significant expense for homeowners.

Many lack personalized data on:

- Solar power generation potential.
- Detailed cost-benefit analysis for solar adoption.

This makes it **difficult to make informed decisions** about investing in solar energy systems.



With energy costs projected to rise 8% this summer, reaching a decade-high average of \$719 per household, the urgency for accessible, personalized solar adoption data has never been greater, as it could empower millions of homeowners to make informed decisions about sustainable energy solutions that significantly reduce their expenses and environmental impact.

# THE SOLUTION

# **SunWise - Smart Solar Savings Predictor**

# FEATURES

**1**

## **Predict Household Electricity Consumption**

Estimate electricity usage and bills based on household attributes to provide a clear view of current energy patterns.

**2**

## **Forecast Solar Power Generation**

Leverage weather-related data to accurately predict solar energy output tailored to individual homes.

**3**

## **Deliver Cost Comparisons**

Compare current electricity costs with potential savings from solar adoption, making the financial benefits tangible.

**THE OUTCOME**

**Enable homeowners to make informed, confident decisions that drive sustainability while achieving long-term cost savings.**



(df-1)

# Household Electricity Bill Data

Leverages monthly electricity bills and household characteristics (e.g., area, rooms, etc.) to estimate and analyze household electricity usage effectively.



## Sample Size

10 columns (attributes) & 1000 rows (entries)

## Attributes

- Number of rooms,
- Number of people,
- House area,
- AC presence,
- TV presence,
- Whether it is a flat,
- Average monthly income,
- Number of children,
- Urban location indicator,
- Amount paid (the monthly bill)

**2****(df-2)**

# Solar Power Generation Data

Historical data from a solar system installed in Berkeley, CA, capturing environmental and time-based factors such as temperature, humidity, and time of day that influence solar power generation.



## Sample Size

16 columns (attributes) & 2920 rows (entries)

## Attributes

- Day of Year, Year, Month, Day,
- First Hour of Period,
- Is Daylight,
- Distance to Solar Noon,
- Average Temperature (Day,)
- Average Wind Direction & Speed (Day),
- Sky Cover,
- Visibility,
- Relative Humidity,
- Average Barometric Pressure (Period),
- Power Generated

**TRANSFORMING RAW DATA INTO ACTIONABLE INSIGHTS**

# DATA CLEANUP

**1**

## **Remove Duplicates**

Identified and removed redundant rows to ensure clean, reliable input.

**2**

## **Handle Missing Values**

### **Numerical Data:**

Filled with column medians to preserve central tendency.

### **Categorical Data:**

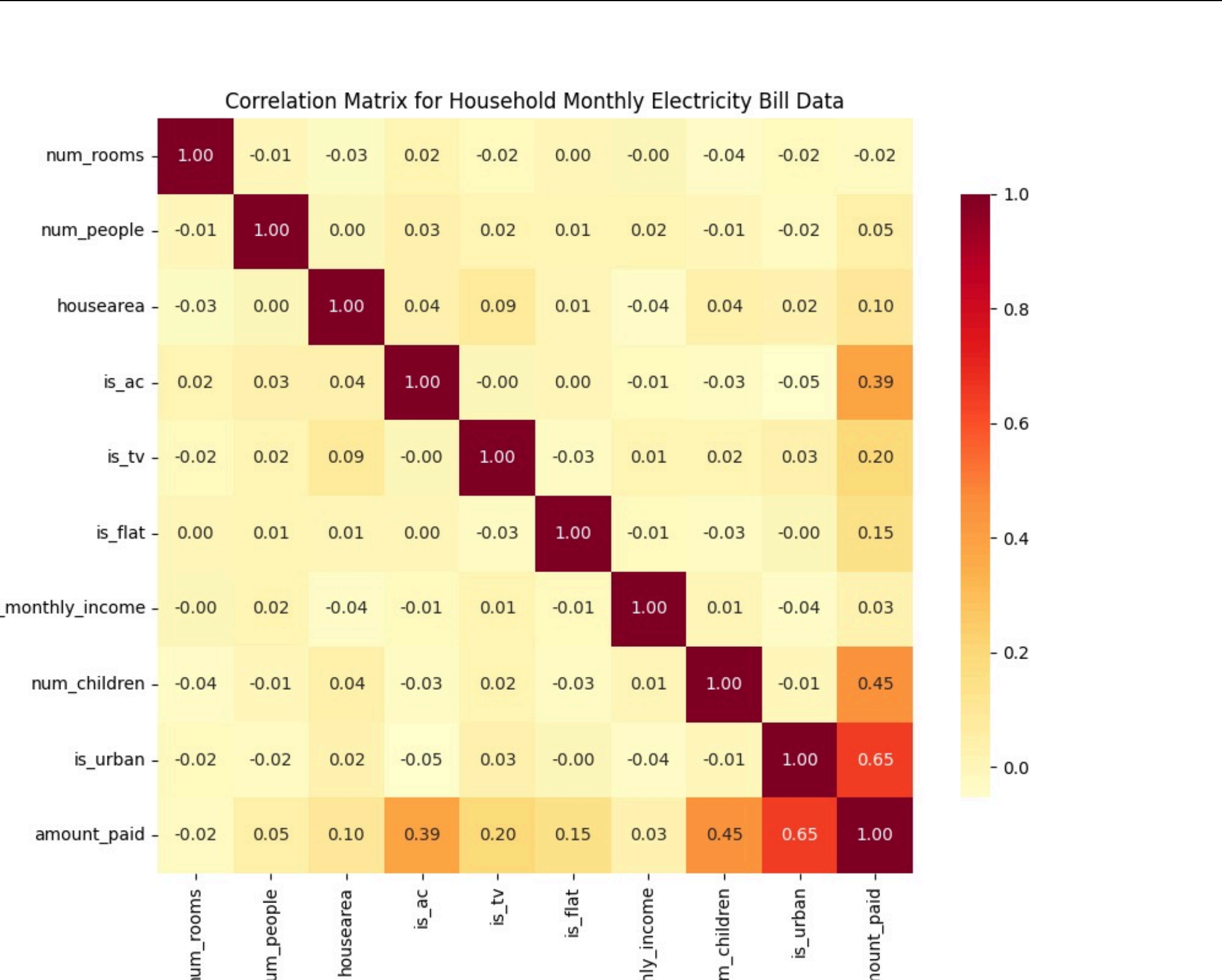
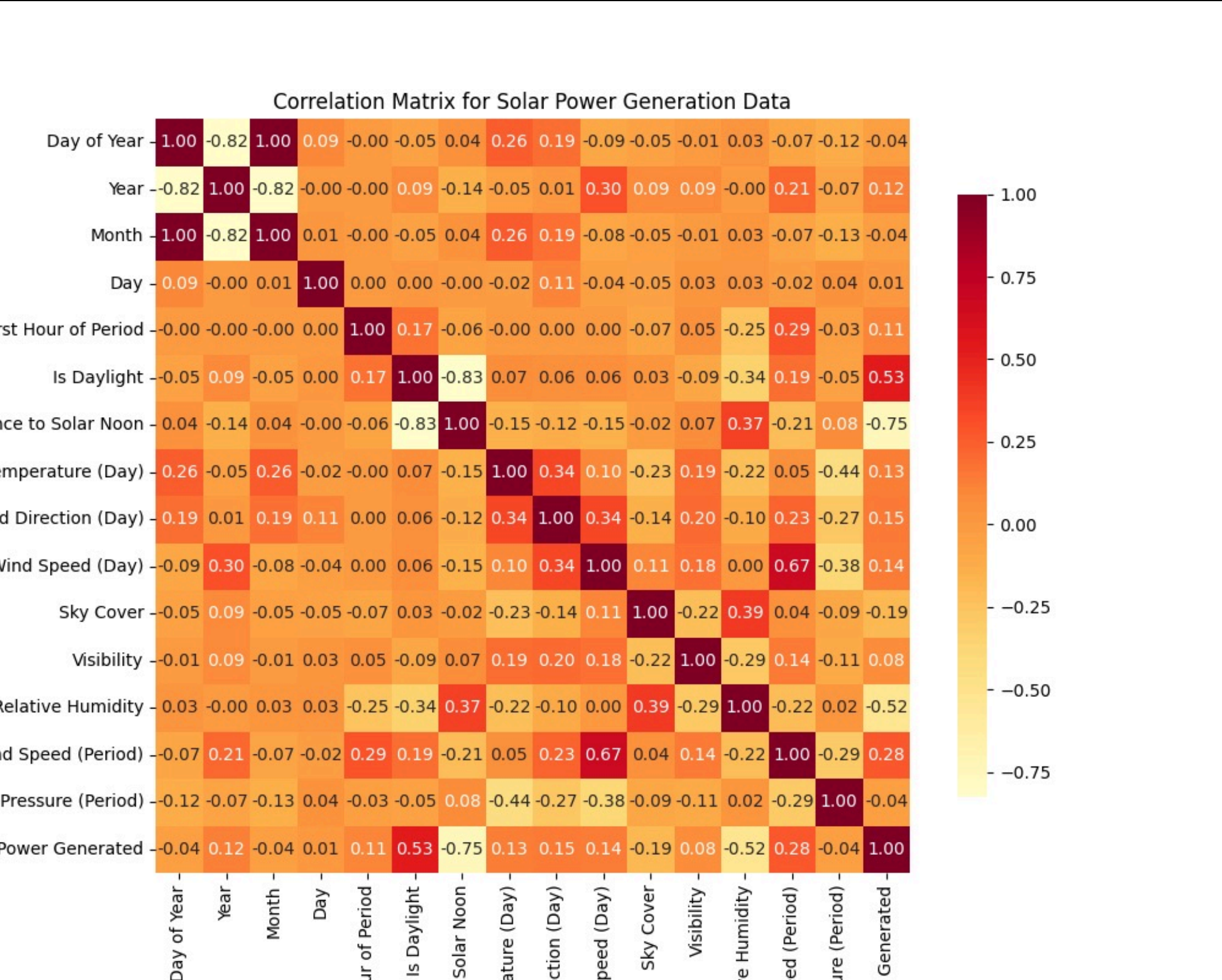
Filled with the most frequent mode value for consistency.

**3**

## **Encode Categorical Variables**

Applied Label Encoding to transform categories into machine-readable integer values.

# CORRELATION ANALYSIS



# THE MODEL

# PREDICTIVE ANALYSIS USING REGRESSION MODELS

## Objective:

- Estimate solar power generation using weather data.
- Forecast household electricity consumption based on household characteristics.

Accurate predictions provide users with a personalized energy profile, empowering them to understand their consumption and potential savings.

## Nature of the Targets

Predicting quantitative outcomes:

- Solar Power Generation (kWh)
- Electricity Consumption (kWh)

Continuous, numeric outputs align naturally with predictive modeling techniques.

## Need for Quantitative Comparisons

Exact energy values enable precise cost-benefit analysis, ensuring users see tangible financial impacts of switching to solar energy.

# MODELS FOR PREDICTING THE AMOUNT OF ELECTRICITY GENERATED

**I**

## Linear Regression

- A simple, interpretable model to analyze relationships between attributes
- Ideal for understanding the baseline trends and impact of individual features.

**2**

## Lasso Regression

- Handles multicollinearity among features.
- Performs automatic feature selection by reducing the weight of less important predictors.
- Enhances performance while preventing overfitting.

**3**

## Random Forest

- Captures nonlinear interactions among household features.
- Handles complex relationships beyond linear assumptions.
- Provides feature importance ranking to reveal the most impactful predictors.

# EVALUATING ALGORITHMS



I have used parameters like **Root Mean Square**,  $R^2$ , **Mean Square Error** and **Mean Absolute Error** to evaluate the estimators



## Linear Regression

- RMSE : 7.864706
- MSE: 61.853594
- MAE: 6.755168
- $R^2$  : 0.884837

## Lasso Regression

- RMSE : 8.504752
- MSE: 72.330799
- MAE: 7.268069
- $R^2$  : 0.865330

## Random Forest

- RMSE : 9.224076
- MSE: 85.083572
- MAE: 7.581606
- $R^2$ : 0.841586

# MODELS FOR PREDICTING THE AMOUNT OF SOLAR POWER GENERATED

**1**

## Random Forest

- Identifies feature importance to highlight influential factors.
- Robust against overfitting, even with high-dimensional data.
- Handles complex relationships without linear assumptions.

**2**

## Gradient Boosting

- Excels at handling noise in data, critical for weather-based predictions.
- Captures non-linear relationships between variables effectively.

**3**

## Artificial Neural Networks

- Learns highly complex, non-linear mappings between features and targets.
- Automatically extracts subtle patterns traditional models may miss.

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# MODEL TRAINING & HYPERPARAMETER TUNING

## Train-Test Split

Separated data into **80% training and 20% testing** for reliable evaluation.

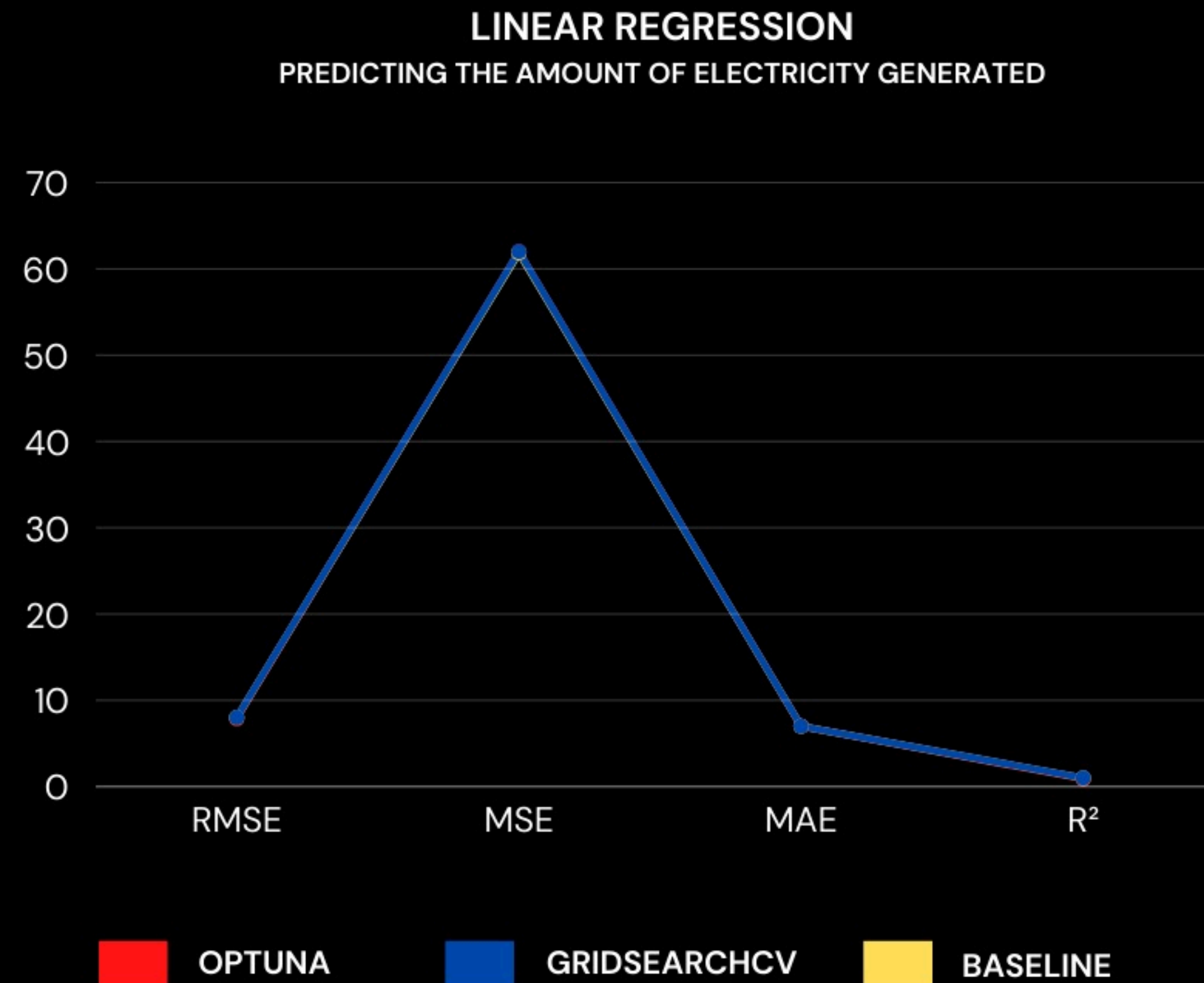
## Hyperparameter Tuning

**Tool Used: Optuna**

### Benefits of Optuna Over Traditional Methods

- Automation: Eliminates the manual effort of tuning hyperparameters.
- Efficiency: Faster and more effective than GridSearchCV, which exhaustively tests all parameter combinations.
- Dynamic Exploration: Optuna intelligently narrows down the search space based on prior results, finding optimal parameters quicker.

# RESULTS OF HYPERPARAMETER TUNING



## RMSE After Optuna:

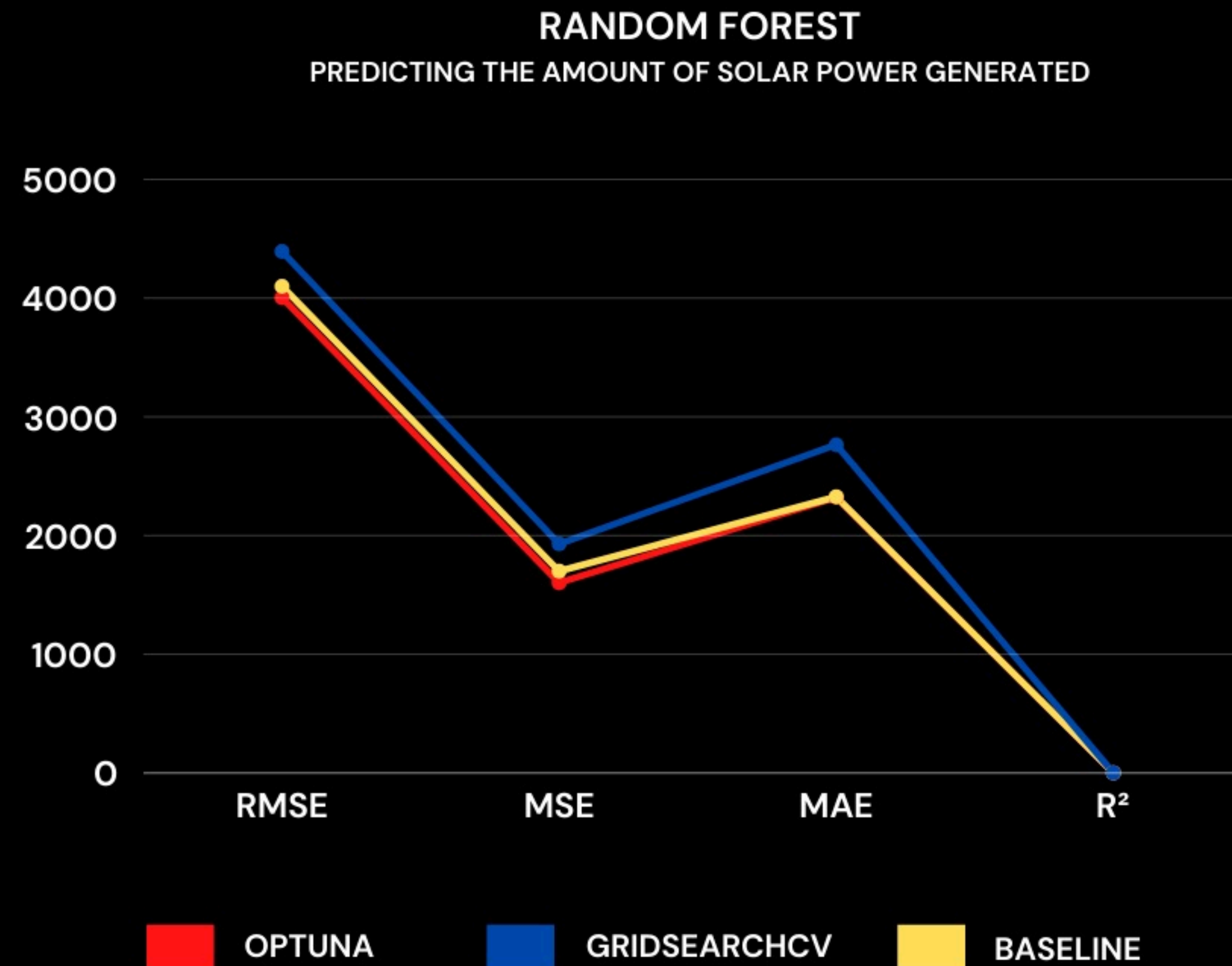
No significant improvement observed

## Insight:

The baseline model performed optimally, as the dataset exhibited a linear relationship between household factors and electricity generation.

Linear Regression remains an effective solution for this task.

# RESULTS OF HYPERPARAMETER TUNING



## RMSE After Optuna:

The RMSE was reduced by approximately 8.87%.

## Insight:

- Optuna outperformed GridSearchCV and baseline methods.
- Highlighted the efficiency of automated tuning in optimizing complex, non-linear models.

# THE RESULTS

# RESULTS

## **Key Outcome:**

- Clear evidence of significant cost savings when switching to solar energy.
- Provides actionable insights on return on investment (ROI) for solar panel installations.
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## **Impact:**

- Highlights how personalized predictions can empower homeowners to make informed, sustainable decisions about their energy usage.